

Executive summary

Uptime Intelligence report: May 2026

Annual outage analysis 2026

Outage prevention remains a central focus for data center operators as demand growth, AI-driven workloads and power constraints reshape risk profiles. Despite — and because of — advances in design and operations, operators are grappling with increasing system complexity, grid instability, growing co-dependencies and evolving external threats. This report analyzes recent data on the causes, frequency and consequences of IT and data center outages.

Uptime Intelligence: actionable insight for the digital infrastructure ecosystem.

To access the entire *Annual outage analysis 2026* report and Uptime Intelligence on an evaluation basis, please visit intelligence.uptimeinstitute.com/request-evaluation

Members of Uptime Network can download the full report on the Network website: network.uptimeinstitute.com

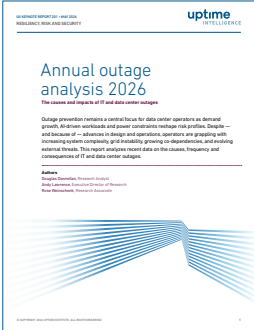
Uptime Intelligence is a research subscription service offered by Uptime Institute. It delivers in-depth, clear analysis and practical guidance focused on the present and future of data center and digital infrastructure strategies, technologies and operations. It serves enterprises that are operating their own digital infrastructure or contracting with third parties; providers of colocation, cloud and other infrastructure-as-a-service offerings; and suppliers of technology and services to all operators of digital infrastructure.

Uptime Institute serves all stakeholders that are responsible for IT service availability through industry-leading standards, education, membership, consulting and award programs delivered to enterprise organizations and third-party operators, manufacturers and providers.

This executive summary summarizes the key findings of Uptime Intelligence's *Annual outage analysis 2026* report, and includes sample pages from the full 29-page report, which is available to Uptime Network members and Uptime Intelligence customers.

Key findings

- For the fifth consecutive year, Uptime Intelligence research suggests outage frequency (on a per-site basis) is declining. However, the pace of improvement has slowed compared with previous years, and around one in 10 still say their last outage had serious or severe impacts.
- External infrastructure failures are becoming more prominent in publicly reported outages. Outages linked to fiber and connectivity issues are rising and are more likely to result in extended disruptions.
- Outage costs continue to rise slowly. More than half (57%) of the respondents to Uptime's 2025 annual survey say their most recent major outage cost more than \$100,000. For the second year in a row, one in five say their most recent impactful outage cost more than \$1 million.
- Power remains the leading cause of impactful outages, but risks are evolving. Failures involving UPS systems, transfer switches and generators are dominant; however, worsening grid constraints and high-density workloads are introducing new pressure points.
- Operators are adapting investment strategies toward automation and control systems to manage complexity, while resiliency assessments remain more focused on internal systems than on external and systemic risks.
- The frequency of major fires at data centers have increased gradually in recent years, with lithium-ion batteries in UPS systems a clear contributing factor. However, this rise in incidents may be partly explained by the rapid build-out of new facilities, rather than the inherent risk of deploying these batteries.



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Outage frequency and severity

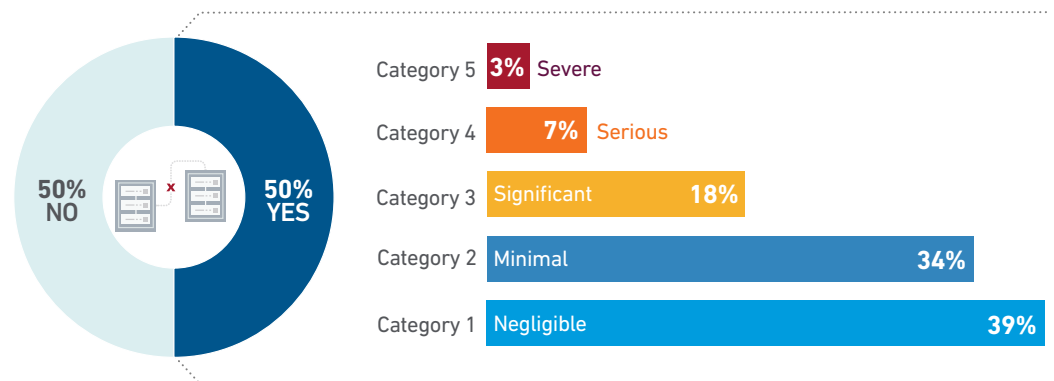
Uptime Intelligence conducted analysis using the latest data from reports and surveys in both 2025 and 2026. While the overall likelihood of experiencing an outage has declined over the past five years, recent results point to a gradual stabilization rather than continued steady improvement.

In the Uptime Institute Global Data Center Survey 2025, half (50%) of operators reported an impactful or serious outage in the past three years (see **Figure 1**). This is the lowest level recorded since 2020 and continues a multi-year trend of improving reliability (see **Figure 2**). However, recent gains have been marginal, suggesting that progress may be slowing. The proportion of incidents classified as serious or severe remains broadly unchanged, at 10% in 2025 compared with 9% in 2024.

Figure 1

One in 20 had a severe or serious outage

Has your organization had an impactful outage in the past three years? If so, how would you classify the most impactful outage on a scale of 1 (negligible) to 5 (severe)? (n=754)



(Responses for "Don't know" are not included.)

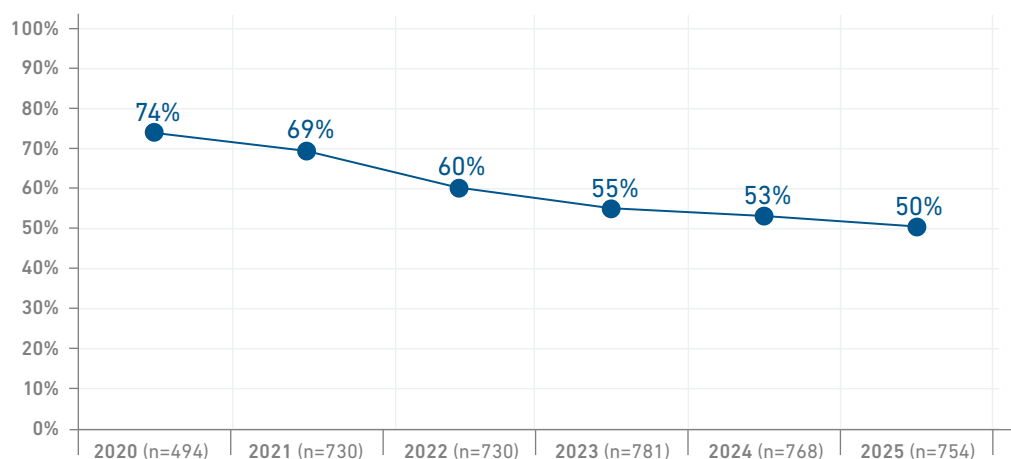
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Figure 2

Gradual improvements in outage frequency continue

Has your organization had an impactful outage in the past three years ("Yes" respondents)



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As businesses and critical services rely on digital infrastructure, even brief outages now affect more users and disrupt more operations, leading to greater financial consequences. Some observations on these trends include:

1. Reliability gains are continuing but diminishing.

Outages are becoming less frequent relative to the overall IT growth, but recent improvements in the outage rate are incremental. This suggests diminishing returns from current resiliency approaches, and that further gains may require changes in design, operations or risk management.

It may also be that improvements in data center practices and design are being offset by countertrends (see below).

2. Stable performance does not indicate lower risk.

The plateau in outage frequency and severity does not suggest operator complacency; instead, resiliency improvements are likely offset by an increasingly complex and demanding operating environment. As data centers support more critical workloads, even isolated incidents can carry broader consequences.

3. System complexity is increasing as resiliency improves.

Software-based resiliency has improved availability for many operators. For example, approaches such as distributed resiliency and availability — which spreads active workloads across several locations — can reduce the impact of a single-site failure. But while these can reduce outage risks caused by single points of failure, they also increase the cost and complexity of synchronizing workloads and may introduce IT-level vulnerabilities that go unnoticed or unforeseen within complex architectures.

4. Power and infrastructure constraints are emerging as systemic risks.

Power availability, interconnection delays and grid reliance are becoming key factors in reliability. With both the grid and the data center operator running much closer to capacity, the likelihood and cost of failures may be greater. While grid failures should not result in a data center outage, transfers to generators sometimes fail.

5. AI workloads are reshaping risk profiles.

AI and high-density compute are placing new stress on power and cooling systems, especially in legacy facilities. Higher rack densities, load variability and operating closer to available power limits may increase the likelihood of cascading failures (see AI data centers and resiliency).

6. Use of substitute components.

Extremely high demand has created a global shortage of critical data center equipment. Developers and operators may face significant delays in securing generators, transformers, switchgear, UPS systems and other subsystems. To work around this, some owners are buying second-user equipment or sourcing substitute components from suppliers without a proven data center track record. This is believed to have contributed to several failures and incidents at some data centers.

Maintaining current levels of reliability may itself become more challenging as demand, density and external dependencies continue to grow. Future improvements will likely depend less on incremental gains in redundancy and more on how effectively operators manage power availability, coordinate across interconnected systems and adapt to more dynamic workloads.

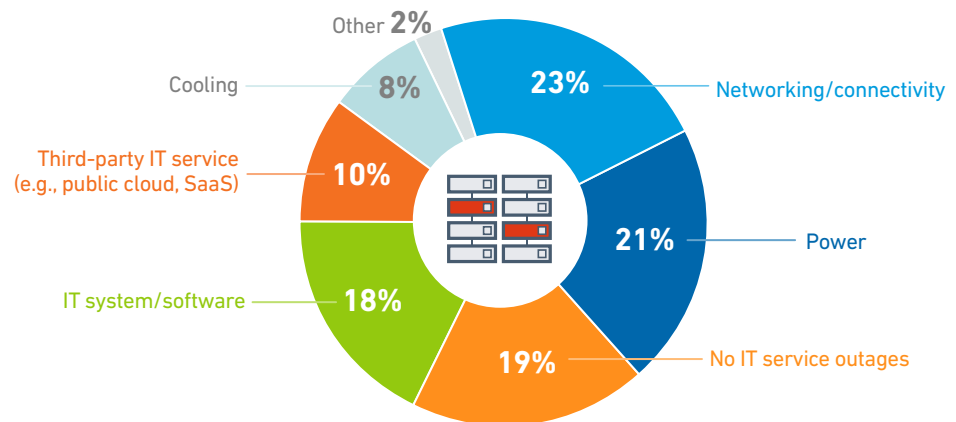
For a broader view of outage causes that extend beyond (but include) the data center, the Uptime Institute Data Center Resiliency Survey 2026 also tracks the most common causes of end-to-end IT service outages, regardless of whether they were the most recent or the most impactful.

Network-related outages continue to be the most frequently reported cause of IT service disruptions (see **Figure 5**), although by a lower margin than in previous years. Notably, nearly one in five respondents report having no IT service outages in the past three years — an improvement of nine percentage points over 2024. This may be partly due to the wider use of software-based strategies, such as automated failover and traffic rerouting, alongside better monitoring and detection tools that help contain issues before they escalate. This is a difficult category of causes to track: some disruptions may also be absorbed within distributed systems, making them less visible as discrete outages.

Figure 5

Most common causes of IT service-related outages

Which of the following issues has been the most common cause of any IT service outages that may have affected your organization over the past three years? (n=403)



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PUBLICLY REPORTED OUTAGES

Commercial operators in the limelight

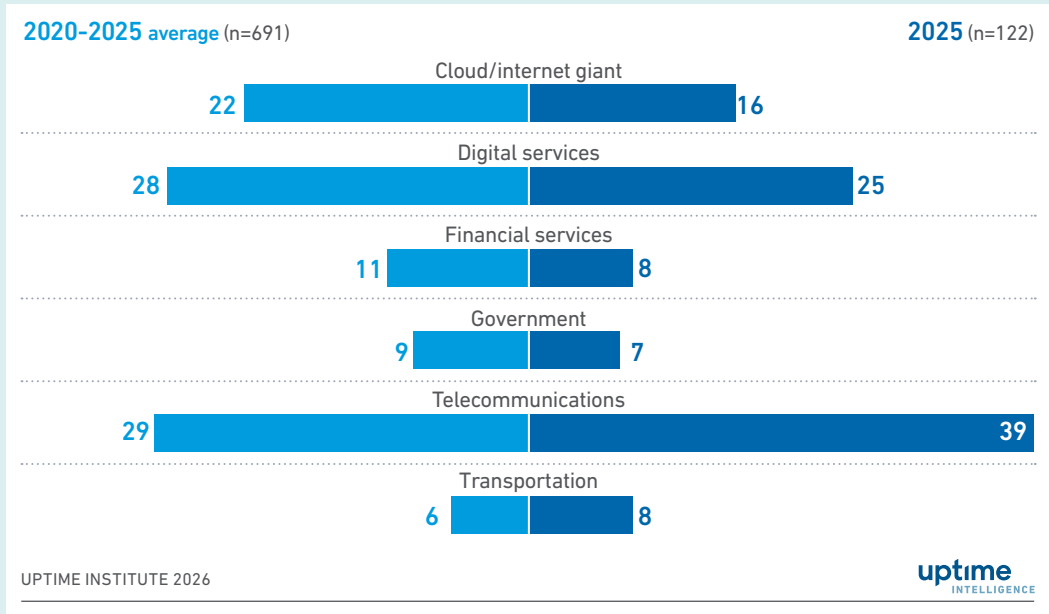
Over 10 years of tracking publicly reported outages, third-party IT and data center providers — including cloud, telecommunications and colocation companies — have consistently accounted for a majority of incidents. This reflects the continued shift toward outsourced services (i.e., the cloud) and distributed digital infrastructure, where large-scale failures can have a widespread impact.

Building on this trend, outages involving telecommunications providers in 2025 rose significantly above the 2020-2025 average (see **Figure 7**). This likely reflects the exposure of external connectivity infrastructure to a broader range of risks, including accidental damage, extreme weather, and, in some regions, geopolitical instability. Telecommunications networks are particularly exposed because they span large geographic areas and depend on infrastructure outside controlled data center environments.

By contrast, outages involving cloud and internet providers fell below average. This likely results from ongoing investments in distributed resiliency, which have strengthened their ability to prevent or recover from disruptions.

Figure 7

Publicly reported outages by sector, 2020 to 2025

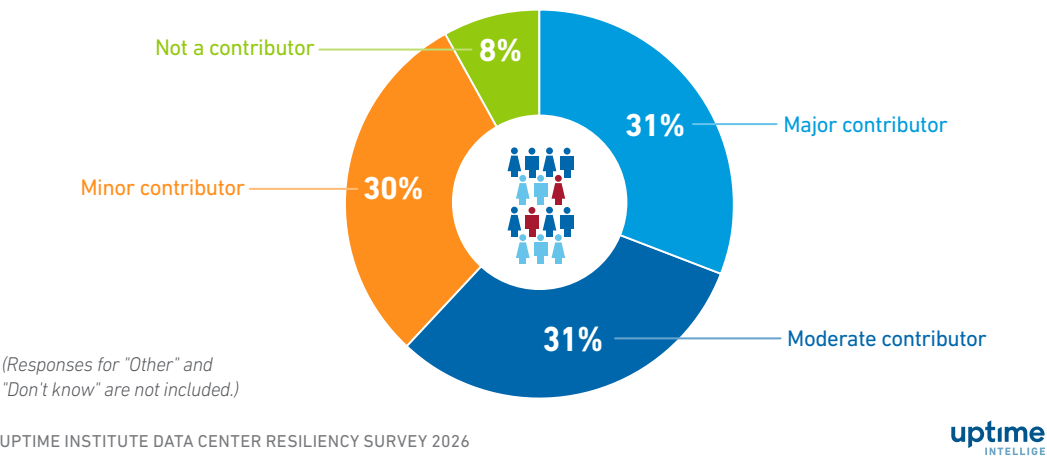


As a result, Uptime generally treats human error as a contributing factor rather than a root cause. Historical data shows that between two-thirds and four-fifths of major IT and data center outages involve some element of human error. Results from the Uptime Institute Data Center Resiliency Survey 2026 affirm this, as 92% of respondents say that human error is at least a minor contributor to their most recent impactful outage (see **Figure 13**). This reinforces the view that even when technical failures are the immediate cause, underlying human and organizational factors are often involved.

Figure 13

Human error contributes to most outages

To what extent would you say human error has contributed to any significant, serious, or severe outage(s) that your organization has experienced over the past three years? (n=220)



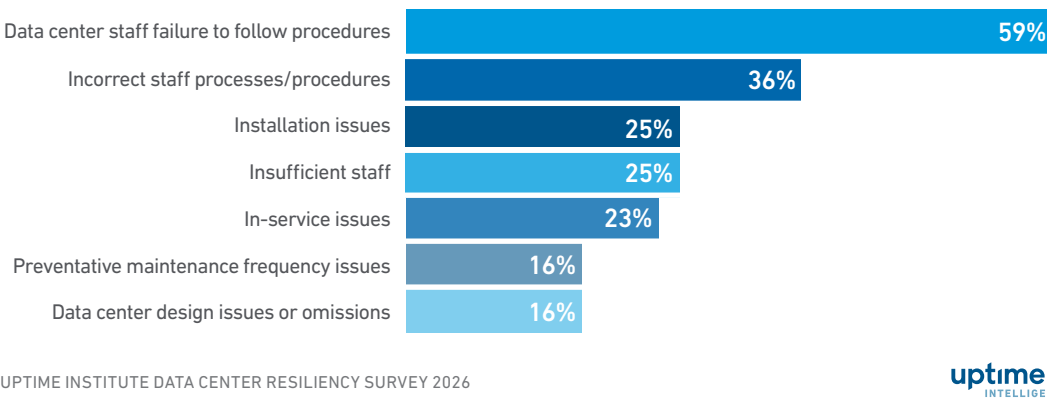
As in previous years, failures to follow established procedures remain the leading driver of human error-related outages (see **Figure 14**). Issues such as inconsistent or unclear processes are also common, alongside installation and in-service errors.

These patterns suggest that managing human error is less about eliminating mistakes and more about designing systems that anticipate and mitigate them. As data center environments become more complex and automated, human operator roles are shifting toward oversight of increasingly dynamic systems, where errors can propagate more quickly and have wider impact.

Figure 14

Most common causes of major human error-related outages

What are the most common ways human error has contributed to IT service outages at your organization's data centers over the past three years? Choose no more than three. (n=199)



Find out more

We hope you found this executive summary of our *Annual outage analysis 2026* report valuable.

To access the entire *Annual outage analysis 2026* report and Uptime Intelligence on an evaluation basis, please visit intelligence.uptimeinstitute.com/request-evaluation

The full report is available to Uptime Network members and Uptime Intelligence subscribers.

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About Uptime Institute

Uptime Institute is the Global Digital Infrastructure Authority. With over 4,000 awards issued in over 122 countries around the globe, and over 1,100 currently active projects in 80+ countries, Uptime has helped tens of thousands of companies optimize critical IT assets while managing costs, resources, and efficiency. For over 30 years, the company has established industry-leading benchmarks for data center performance, resilience, sustainability, and efficiency, which provide customers assurance that their digital infrastructure can perform across a wide array of operating conditions at a level consistent with their individual business needs. Uptime's Tier Standard is the IT industry's most trusted and adopted global standard for the design, construction, and operation of data centers. Offerings include the organization's Tier Standard and Certifications, Management & Operations reviews and assessments including SCIRA-FSI financial sector risk assessment, the Sustainability Assessment, and a broad range of additional risk management, performance, availability, and related offerings. Uptime Education training programs have been successfully completed by over 100,000 data center professionals, such as the much-valued ATD (Accredited Tier Designer) and AOS (Accredited Operations Specialist). The Uptime Education curriculum has been expanded by the acquisition of CNet Training Ltd. In 2023.

Uptime Institute is headquartered in New York, NY, with offices in London, Sao Paulo, Dubai, Riyadh, and Singapore, and full-time Uptime professionals based in over thirty-four countries around the world.

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